ZCAS University

CCS1501 ENGINEERING MATHEMATICS

MID-SEMESTER EXAMINATION

WEDNESDAY, 22ND APRIL, 2026

TIME: 13:00 TO 14:30 HOURS

TIME ALLOWED: THREE HOURS (plus 5 minutes to read through the paper)

INSTRUCTIONS:

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1. This question paper consists of **three (03)** questions. You are required to answer **ANY TWO** questions.
2. Cellular phones are **not allowed** in the examination room.
3. Electronic calculators are **NOT** allowed in the examination room.
4. Candidates must **not turn this page** until the invigilator tells them to do so.

QUESTION ONE

(a) Write down the first three terms in the Binomial expansion of $\sqrt{1+x}$ **[3 Marks]**

(b) Given the polynomial $P(x) = x^3 + 4x^2 + x - 6$

(i) Factorize $P(x)$ completely **[3 Marks]**

(ii) Hence, resolve $\frac{5x+19}{x^3+4x^2+x-6}$ into partial fractions **[4 Marks]**

(c) Given that $f(x) = ax^3 + (a+b)x^2 + (a+2b)x + 1$ is exactly divisible by $(x+1)$.

(i) Express b in terms a. **[3 Marks]**

(ii) Use synthetic division to find the **quotient** when $f(x)$ is divided by $(x+1)$. **[3 Marks]**

(d) Find the value of n if the coefficient of the second term in the expansion of $(1-x)(1+2x)^n$ is 19. **[4 Marks]**

[Total: 20 Marks]

QUESTION TWO

(a) Given the equation $ax^2 + bx + c = 0$, show that $x = \frac{-b \pm \sqrt{b^2-4ac}}{2a}$ **[5 Marks]**

(b) If α *and* β are roots to the equation $2x^2 - 3x + 2 = 0$, find the quadratic equation whose roots are $\frac{1}{\alpha\beta}$ *and* $\frac{1}{\alpha+\beta}$. **[3 Marks]**

(c) When the expression $x^3 + ax^2 - (ax^2 + 12)x + 7a + 10$ is divided by $x - 1$ the remainder is 7.

(i) Find the value of a **[3 Marks]**

(ii) Hence, factorize the expression completely. **[5 Marks]**

(d) Find the 5th term of the expansion $\left(x + \frac{2}{x}\right)^{10}$ **[4 Marks]**

[Total: 20 Marks]

QUESTION THREE

(a) (i) Define a one-to-one function **[2 Marks]**

(ii) Show that $f(x) = 9x - 2$ is a one to one function **[3 Marks]**

(b) Given that $f(x) = 2x^3 - x^2 - 5x - 2$

(i) Show that $f(x)$ is divisible by $x - 2$. **[2 Marks]**

(ii) Hence, solve the inequality $f(x) \leq 0$ **[5 Marks]**

(c) (i) Express as a single logarithm $2\log_4(2x + 3) - \log_4 x - \log_4(2x - 1)$ **[1 Mark]**

(ii) Hence, solve the equation $2\log_4(2x + 3) = 1 + \log_4 x + \log_4(2x - 1)$ **[3 Marks]**

(d) Solve the equation $4(3^{2x+1}) + 17(3^x) - 7 = 0$ **[4 Marks]**

[Total: 20 Marks]

END OF EXAMINATION.